

Customer Review Clustering for Marketing Insights Using KMeans

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ABSTRACT

In the current digital era, customer feedback acts as a valuable resource for companies looking to enhance their products and promotional tactics. However the vast availability of unstructured reviews make manual analysis in traditional method inefficient and less accurate. This project leverages machine learning, specifically clustering algorithms, to analyze Amazon product reviews.

The main objective is to categorize customers according to their review behavior to discover significant patterns and insights. Utilizing unsupervised learning methods on structured review data, this study reveals various customer personas and suggests customized marketing strategies for every segment.

The project segments customers into meaningful groups and provides distinct feedback patterns for each segment. The result affirms better engagement, retention and strategic planning based on data informed approach.

Keywords : Customer Segmentation , KMeans Clustering , Unsupervised Learning , Amazon product review , Marketing Analysis , Customer Insights , Behavioral Data Analysis.

INTRODUCTION

Analyzing customer feedback on a large scale is essential for making well-informed business decisions. Market segmentation traditionally are based on demographics, psychographics or purchase history. However, with digital platform review behaviour itself becomes rich dimension for market analysis.

Clustering algorithms like KMeans can automatically categorize customers who exhibit similar behaviors, facilitating targeted actions for specific segments. The approach involves comprehensive data processing , enhanced feature engineering and internal validation using evaluation metrics.

In this project, we process the Amazon review dataset, create new features that reflect customer sentiment and behavior, and apply clustering techniques to identify and examine customer segments. The ultimate goal is to translate these clusters into marketing insights. Each cluster is analyzed to propose targeted strategies.

This approach moves beyond simple review rating analysis and dives deep into the behavioral market analysis, demonstrating the power of machine learning in business applications.

RELATED WORKS

Several notable works in the field of e-commerce and market analysis that I referred while working on this project have laid groundwork for applying unsupervised learning to consumer behavior modelling.

Jain and Ahuja(2014) developed a KMeans based clustering model to categorize online customers into distinct profiles. Their work emphasized on understanding pre-purchase behavior pattern for customer engagement.

While their work was grounded in transaction data our project extends behavioural segmentation focusing on engagement with review content, helpfulness and sentiment indicators.

Li and Lee(2024) did a comparative study between KMeans, SVM and KMA and found out KMeans outperform SVM. This comparison was quite relevant to this project in order to choose the best approach while moving forward and yielded some valuable insights.

Jain (2010) talked about practical advantages of KMeans in customer analytics. Their views align with this project's choice of KMeans over any other models hence enabling clear segment profiling for actionable market decisions.

Han, Kamber and Pei(2011) provided the explanation of validation metrics such as Silhouette Score and Calinski-Harabasz Index are used in this project to determine optimal cluster.

LITERATURE REVIEW

In her paper Jain emphasized on creating a model that attempts to study the segmented profile of the online consumers in the context of their pre purchase behaviour(2014). Their work utilized Kmeans clustering to segment online users on e-commerce platform, categorizing them into window shoppers, loyal buyers and infrequent users. Jain's study validated the core idea of using unsupervised learning for behavioural segmentation.

Li and Lee(2024)'s research integrates clustering algorithms within a big data ecosystem. Their model is capable of real-time adjustment to evolving customer behavior. While talking about comparison of accuracy provided by KMeans and Svm Xaotong Li and Lee.(2024) mentioned that while SVM provides 72.36% accuracy KMeans provides 81.08% accuracy.

In the textbook "Data Mining : Concepts and Techniques" Han, Kamber and Pei(2011) defines the theoretical foundation for Kmeans clustering and cluster validation metrics. The use of internal evaluation tools such as silhouette score and calinski-harabasz index in this project directly follows their recommended practices.

Jain (2010) stated in his paper that KMeans is a reliable algorithm for market segmentation particularly when interpretability and computational efficiency are priorities.

PROBLEM STATEMENT

This project aims to address the problem

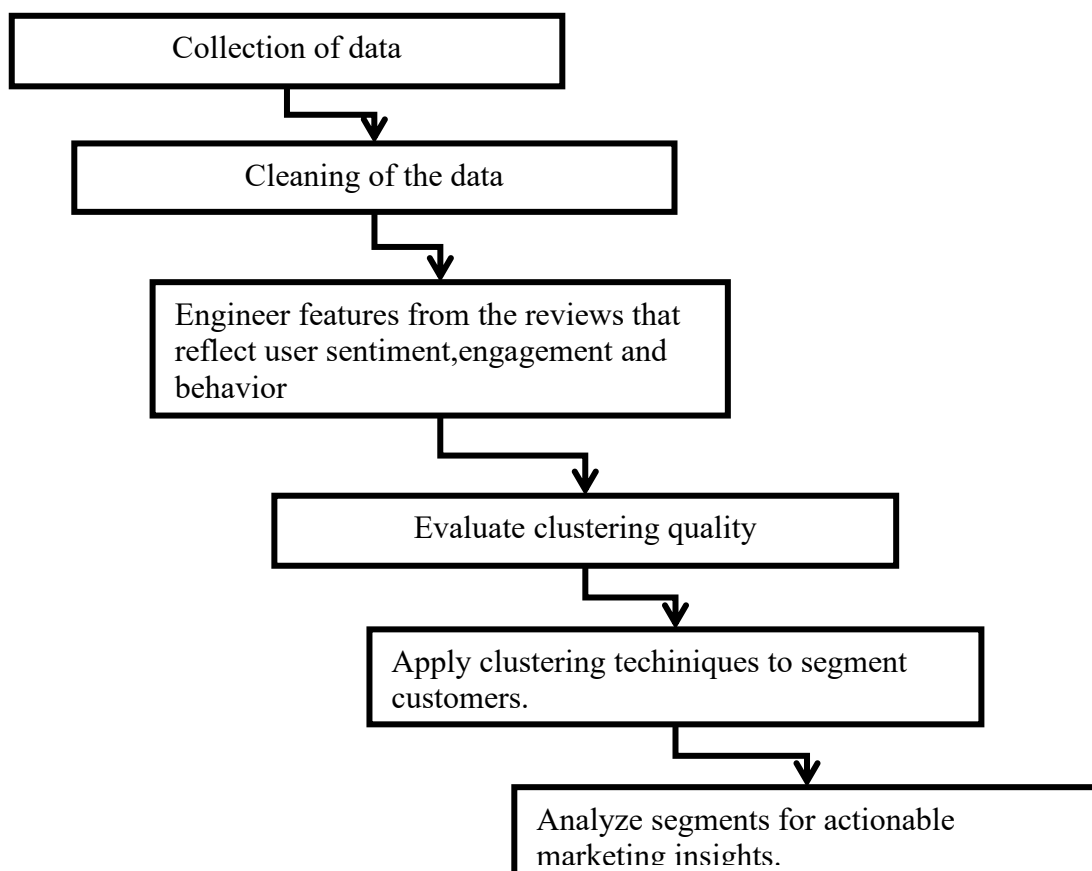
How can we use unsupervised clustering algorithm to segment customers based on their review behaviour and how can these segments be translated into marketing strategies?

By focusing on clustering Amazon product review data, the project seeks to go beyond traditional analysis methods and develop a data driven approach for behavioural segmentation.

OBJECTIVE

1. Identifying features that represent customer behaviour and engagement best in the data.
2. Choosing appropriate clustering method
3. Validating the quality of clusters
4. Analyzing the behavioural qualities of each segment to recommend personalized marketing approach.

WORK FLOW



DATA COLLECTION

Resource

Dataset: Amazom Product Review Dataset
Source: Kaggle
Size : 843 X 19(post formating)
843 X 27(actual dataset)
Datatype: CSV format

id	Unique identifier for the product
asins	Amazon Standard Identification Number
brand	Brand of the product
name	Product name
categories	Product category
colors	Color of the product
dateAdded	Date the product was added to the dataset
dateUpdated	Date the product metadata was updated
manufacturer	Manufacturer name
prices	Price of the product
reviews.date	Date the review was posted
reviews.rating	Customer rating
reviews.text	Full textual review
reviews.title	Title of the review
reviews.username	Username of the reviewer
reviews.numHelpful	Number of user marked it helpful
reviews.doRecommend	Wheter the user recommends the product or not
reviews.sourceURLs	URL pointing to the original review
keys	Product key information

DATA PREPROCESSING

1. Handeling missing values:

Categorical columns like colors filled with 'Unknown'

Numerical columns like reviews.rating filled with the mean

```
# Fill categorical columns
dataset['colors'] = dataset['colors'].fillna('Unknown')
dataset['reviews.doRecommend'] = dataset['reviews.doRecommend'].fillna('Unknown')
dataset['reviews.title'] = dataset['reviews.title'].fillna('No Title')
dataset['reviews.username'] = dataset['reviews.username'].fillna('Anonymous')

# Fill numerical columns
dataset['reviews.numHelpful'] = dataset['reviews.numHelpful'].fillna(0)
dataset['reviews.rating'] = dataset['reviews.rating'].fillna(dataset['reviews.rating'].mean())
```

2. Feature Scaling(Standardization):

$Z = \frac{\text{value} - \text{mean}}{\text{standard deviation}}$

standard deviation

applied to normalize features before clustering

```
# Scale features for better clustering
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

3. **Feature Engineering** : Created new columns such as review_length, helpfulness_ratio, is_high_rating, engagement_score, rating_deviation.

```
# Enhanced feature engineering
def create_enhanced_features(df: Any):

    # Make a copy to avoid warnings
    df = df.copy()

    # Basic review characteristics
    df['review_length'] = df['reviews.text'].str.len()
    df['word_count'] = df['reviews.text'].str.split().str.len()

    # Rating-based features
    df['rating_deviation'] = df['reviews.rating'] - df['reviews.rating'].median()
    df['is_extreme_rating'] = ((df['reviews.rating'] <= 2) | (df['reviews.rating'] >= 5)).astype(int)
    df['is_high_rating'] = (df['reviews.rating'] >= 4).astype(int)
    df['is_low_rating'] = (df['reviews.rating'] <= 2).astype(int)

    # Helpfulness features
    df['helpfulness_ratio'] = df['reviews.numHelpful'] / (df['reviews.numHelpful'].max() + 1)
    df['is_helpful'] = (df['reviews.numHelpful'] > df['reviews.numHelpful'].median()).astype(int)
    df['helpfulness_per_char'] = df['reviews.numHelpful'] / (df['review_length'] + 1)

    # Review engagement features
    df['words_per_rating'] = df['word_count'] / df['reviews.rating']
    df['engagement_score'] = df['reviews.numHelpful'] * df['reviews.rating']

    # Product-based features
    df['product_category'] = pd.Categorical(df['name']).codes
    df['brand_category'] = pd.Categorical(df['brand']).codes

    # Time-based features
    if 'reviews.date' in df.columns:
        df['reviews.date'] = pd.to_datetime(df['reviews.date'], errors='coerce')
        df['review_year'] = df['reviews.date'].dt.year
        df['review_month'] = df['reviews.date'].dt.month
        df['days_since_min'] = (df['reviews.date'] - df['reviews.date'].min()).dt.days

    # Text length categories
    df['review_length_category'] = pd.cut(df['review_length'],
                                         bins=[0, 100, 300, 1000, float('inf')],
                                         labels=['Short', 'Medium', 'Long', 'Very_Long'])
    df['review_length_cat_code'] = pd.Categorical(df['review_length_category']).codes

    # Word density features
    df['words_per_char'] = df['word_count'] / (df['review_length'] + 1)

    return df
```

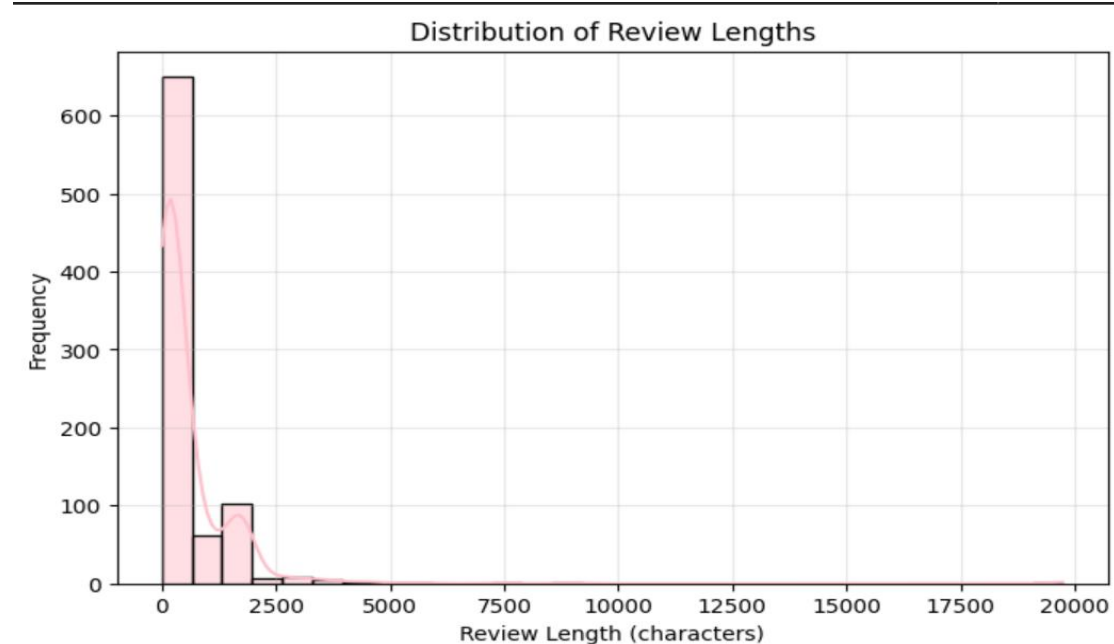
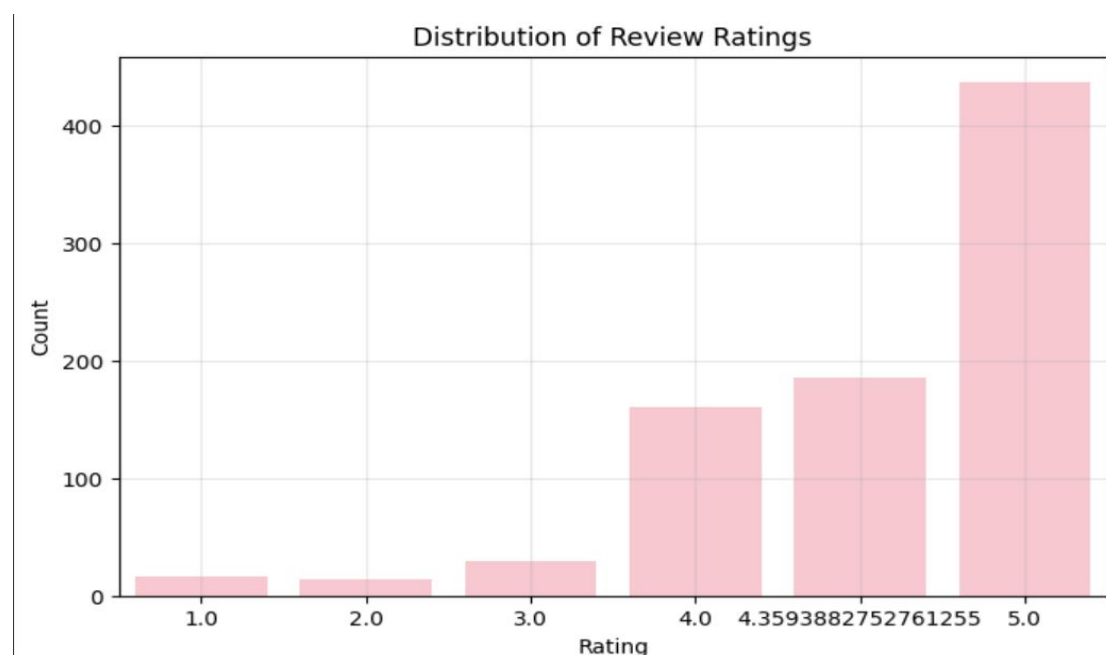
4. **Dropped Columns** : dimension, ean, manufacturerNumber, upc, weight were dropped as they were not critical for analysis.

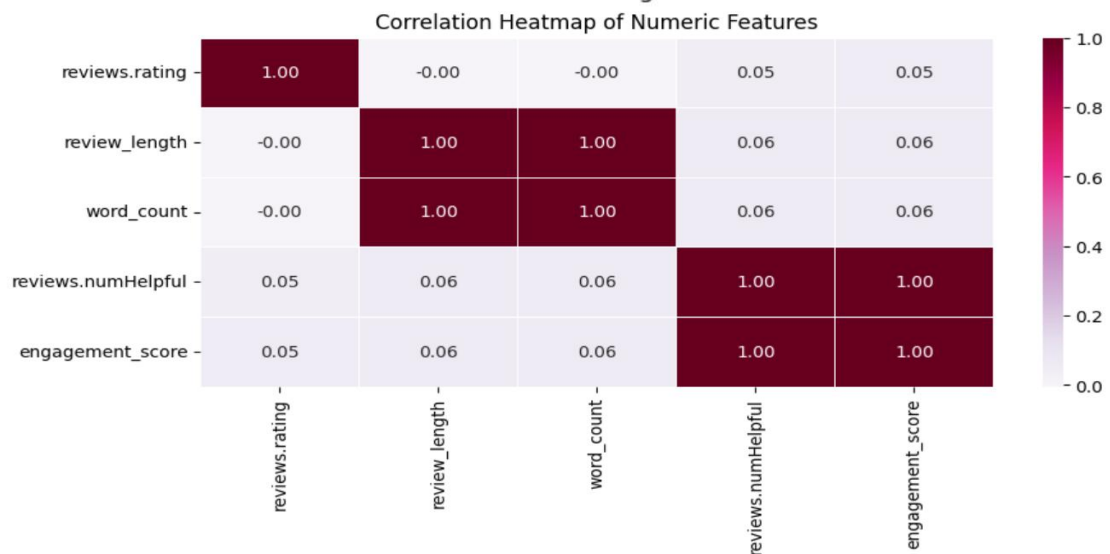
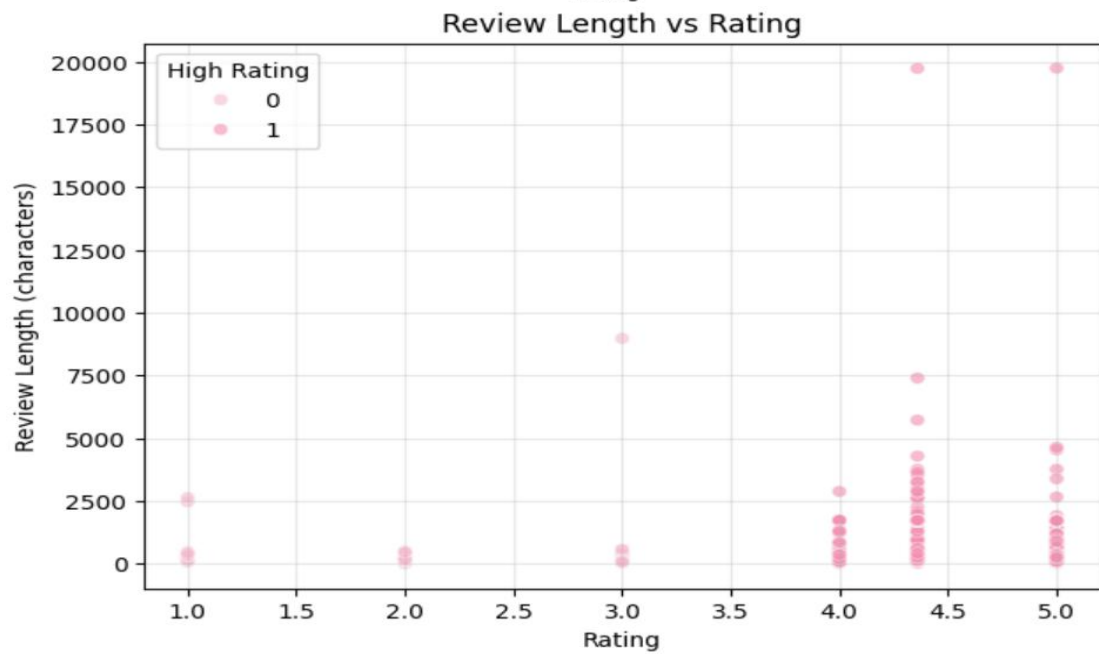
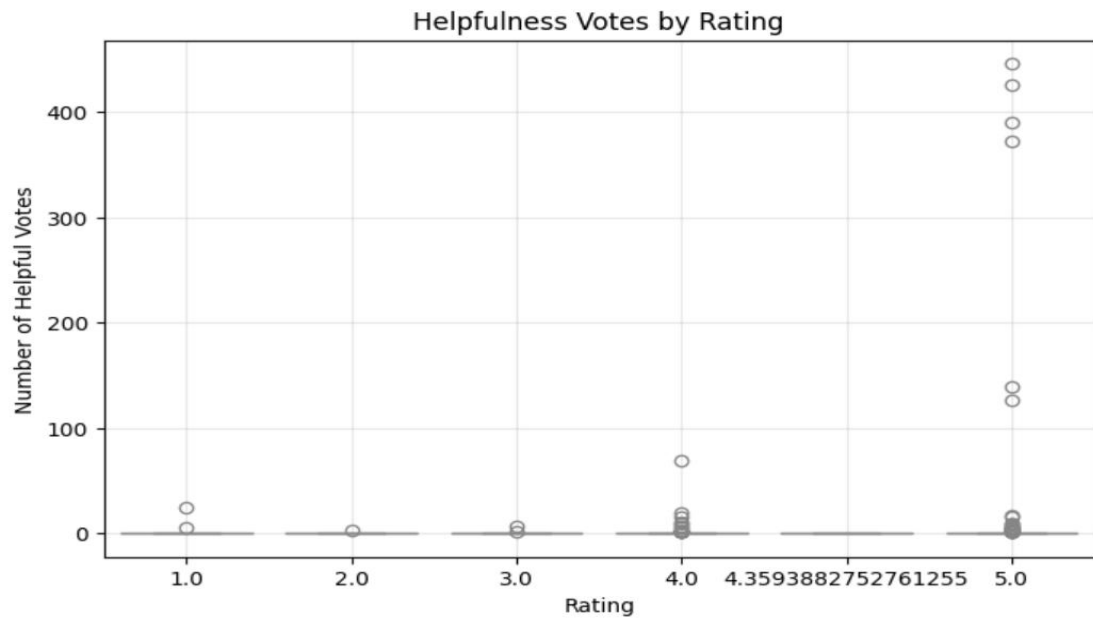
```
# Drop columns not critical to analysis
dataset = dataset.drop(columns=['dimension', 'ean', 'manufacturerNumber', 'upc', 'weight'], errors='ignore')
dataset
```

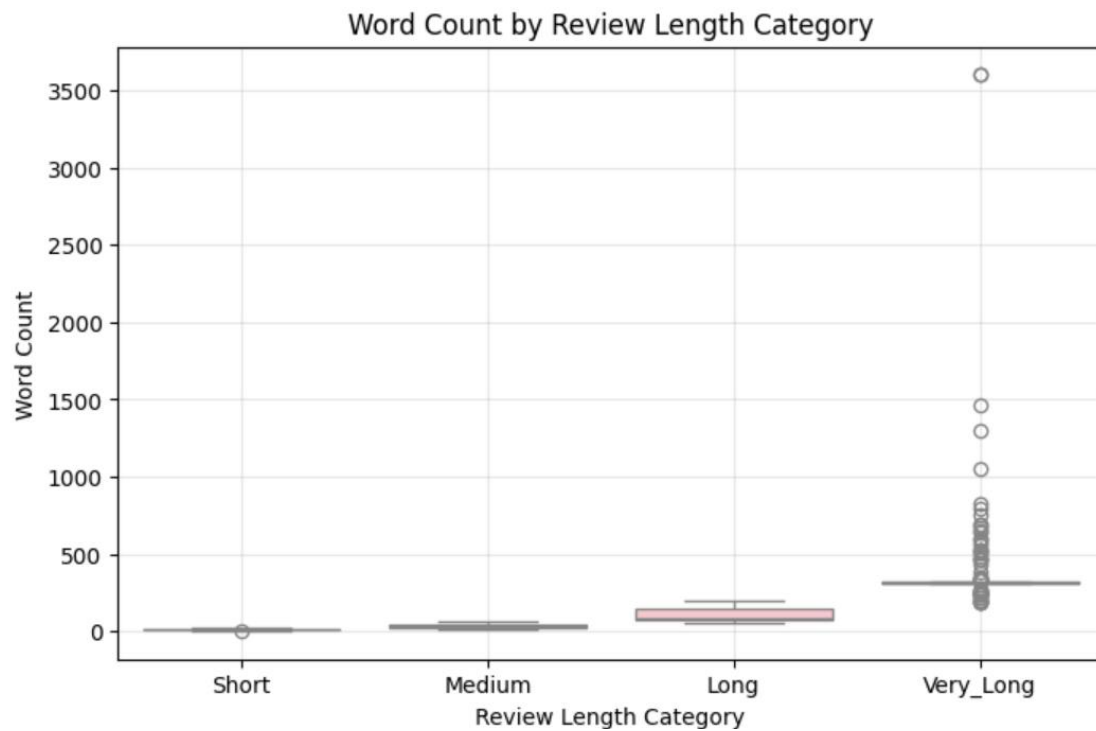
EXPLORATORY DATA ANALYSIS

EDA involves visual and statistical analysis to understand data distribution relationship.

1. Most of the reviews are 4 to 5 star reviews indicating positive customer sentiment overall.
2. Detailed reviews tend to receive more upvotes suggesting review length can also be a factor of engagement.
3. High engagement score points to customer who are both positive and impactful.
4. Plots showing that textual review patterns vary across rating level such as low rating reviews often have high word count suggesting dissatisfied customers explain their issues in more depth and it is also considered helpful for others.
5. Correlation heatmaps revealed that engagement_score and review_rating are positively correlated with helpfulness votes highlighting key features for clustering.







BUISNESS INSIGHTS ANALYSIS

1. Customers with consistent 5-star ratings are ideal for loyalty programs.
2. Early detection of dissatisfaction allows timely intervention.
3. Detailed reviews often provide valuable feedback and may influence others.
4. Loyal customers but they keep the review short needed to be encouraged for detailed feedback.
5. Some product have dedicated loyal segments.

ML MODEL CREATION

Model used : **KMeans Clustering**

KMeans is an unsupervised machine learning algorithm used to group data into k distinct clusters based on feature similarity. It tries to minimize the variance within each cluster and maximize the variance between clusters.

The objective function

$$J = \sum_{j=1}^k \sum_{x_i \in C_j} ||x_i - p_j||^2$$

where:

$X = \{x_1, x_2, \dots, x_n\}$ be the data point

The centroid of the cluster j is p_j

The set of points assigned to cluster j is C_j

```
# Perform clustering with optimal k
print(f"Performing K-means clustering with k={optimal_k}...")
kmeans_optimal = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
cluster_labels = kmeans_optimal.fit_predict(X_scaled)
```

```
Performing K-means clustering with k=9...
```

Validation Metrics

1. **Silhouette Score** : The metric used to evaluate the quality of clusters in clustering algorithm.

The Silhouette Score for a single sample is :

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where :

$a(i)$ = average distance of point i to all other points in the same cluster

$b(i)$ = lowest average distance of point i to points in other clusters

Average Silhouette Score over all points gives an overall measure of clustering quality.

- Closer to 1 = better-defined clusters.
- Closer to 0 = overlapping clusters.
- Negative = incorrect clustering.

```
# Silhouette score
axes[0].plot(K, sil_scores, 'bo-', linewidth=2, markersize=8)
axes[0].set_xlabel('Number of Clusters (k)')
axes[0].set_ylabel('Silhouette Score')
axes[0].set_title('Silhouette Score vs Number of Clusters')
axes[0].grid(True, alpha=0.3)
best_sil_k = K[np.argmax(sil_scores)]
axes[0].axvline(x=best_sil_k, color='red', linestyle='--', alpha=0.7)
axes[0].text(best_sil_k, max(sil_scores), f'Best k={best_sil_k}', ha='center')
```

2. **Calinski - Harabasz Index** : Evaluation metric used to measure the quality of clustering. It assesses how well separated and dense the clusters are.

$$\text{CH index} = \frac{\text{Tr}(\text{SSB})}{\text{Tr}(\text{SSW})} \cdot \frac{n - k}{k - 1}$$

where :

k = number of clusters

n = number of data points

SSB = between cluster dispersion i.e. how far clusters are from the overall centroid

SSW = within cluster dispersion i.e. how tight the points are in each cluster

Tr = trace of a matrix i.e. sum of diagonal elements.

Higher value indicated better clustering.

```
# Calinski-Harabasz score
axes[1].plot(K, cal_scores, 'ro-', linewidth=2, markersize=8)
axes[1].set_xlabel('Number of Clusters (k)')
axes[1].set_ylabel('Calinski-Harabasz Score')
axes[1].set_title('Calinski-Harabasz Score vs Number of Clusters')
axes[1].grid(True, alpha=0.3)
```

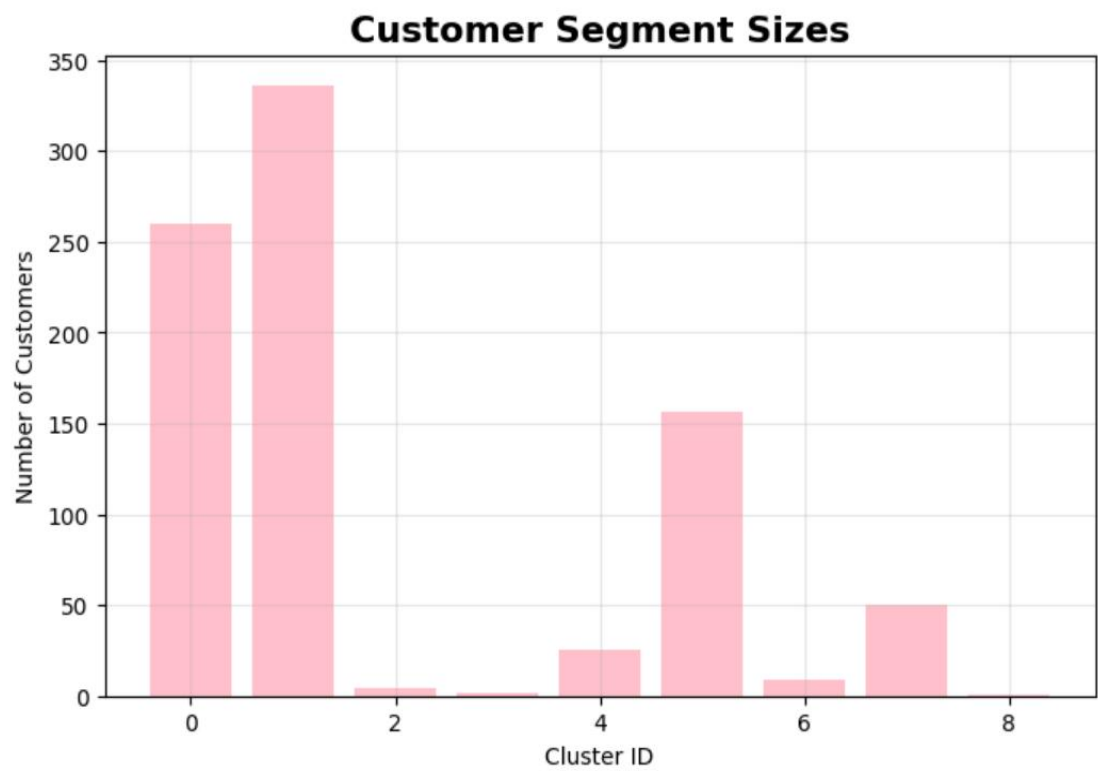
3. **Elbow - Method** : Visual technique used within clustering algorithms like KMeans to determine the optimal number of clusters by analyzing the inertia which is within cluster sum of squared errors.

1. Run KMeans from the range of values of k
2. Calculate Inertia (sum of squared distances from each point to its assigned cluster centroid)
3. Plot the number of clusters k against inertia
4. Look for an “elbow” point- the point after which the inertia reduction slows down.

```
# Elbow method
axes[2].plot(K, inertias, 'go-', linewidth=2, markersize=8)
axes[2].set_xlabel('Number of Clusters (k)')
axes[2].set_ylabel('Inertia')
axes[2].set_title('Elbow Method For Optimal k')
axes[2].grid(True, alpha=0.3)
```

VISUAL PLOTS:

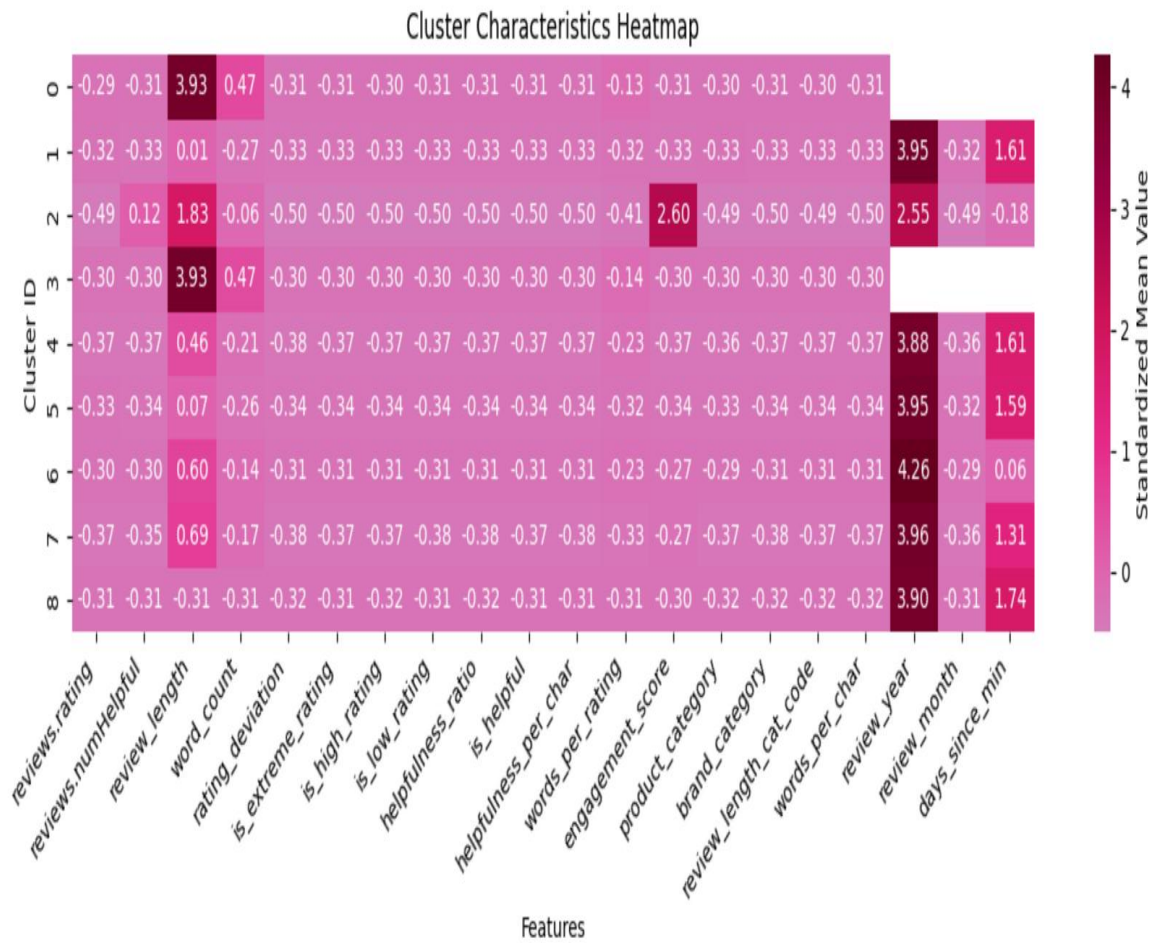
1. Cluster size distribution



2. Rating vs Helpfulness



3.Feature importance map



RESULT AND EVALUATION :

- Using a combination of Silhouette analysis and elbow method the optimal clusters are found to be 9. Thus **k = 9**.
- Silhouette Score = 0.398 indicating moderate cluster separation.
- Calinski - Harabasz is 283.66 indication that higher value means well separated clusters with tight internal cohesion
- Each cluster was analyzed based on its
 1. Average rating
 2. Review length and helpfulness
 3. Top reviewed products
 4. Customer engagement patterns
- Segment Type identified
 1. Higer ratings, constant praise ,ideal for analysis
 2. Positive but less engaged, suitable for upselling a product
 3. Frequent low ratings which require retention efforts
 4. Indifferent behaviour but can be helpful for engagement campaigns
- Each clusters directly informed specific marketing strategies.

----- CUSTOMER SEGMENT ANALYSIS FOR MARKETING -----

--- CLUSTER 0: 260 customers (30.8%) ---

SATISFACTION METRICS:

Average Rating: 4.47 ± 0.31

High Ratings (4-5): 99.6%

Low Ratings (1-2): 0.0%

ENGAGEMENT METRICS:

Average Helpfulness: 0.0

Average Review Length: 1235 characters

Average Word Count: 227 words

TOP PRODUCTS:

1. Amazon Fire TV: 144 reviews (55.4%)

2. Amazon Premium Headphones: 71 reviews (27.3%)

3. Certified Refurbished Amazon Fire TV (Previous Generation - 1st): 26 reviews (10.0%)

SEGMENT TYPE: SATISFIED CUSTOMERS

MARKETING FOCUS: Upsell opportunities, loyalty programs

ENGAGEMENT: Low Engagement, Detailed Reviewers

--- CLUSTER 1: 336 customers (39.9%) ---

SATISFACTION METRICS:

Average Rating: 5.00 ± 0.00

High Ratings (4-5): 100.0%

Low Ratings (1-2): 0.0%

ENGAGEMENT METRICS:

Average Helpfulness: 0.0

Average Review Length: 162 characters

Average Word Count: 31 words

TOP PRODUCTS:

1. Amazon Tap - Alexa-Enabled Portable Bluetooth Speaker: 331 reviews (98.5%)

2. Amazon 5W USB Official OEM Charger and Power Adapter for Fire Tablets and Kindle eReaders: 4 reviews (1.2%)

3. Amazon Fire TV: 1 reviews (0.3%)

SEGMENT TYPE: BRAND ADVOCATES

MARKETING FOCUS: Leverage for testimonials, referrals, premium products

ENGAGEMENT: Low Engagement, Brief Reviewers

--- CLUSTER 2: 4 customers (0.5%) ---

SATISFACTION METRICS:

Average Rating: 5.00 ± 0.00

High Ratings (4-5): 100.0%

Low Ratings (1-2): 0.0%

ENGAGEMENT METRICS:

Average Helpfulness: 408.5

Average Review Length: 1534 characters

Average Word Count: 289 words

TOP PRODUCTS:

1. Amazon Premium Headphones: 3 reviews (75.0%)

2. Amazon Tap - Alexa-Enabled Portable Bluetooth Speaker: 1 reviews (25.0%)

SEGMENT TYPE: BRAND ADVOCATES

MARKETING FOCUS: Leverage for testimonials, referrals, premium products

ENGAGEMENT: High Engagement, Detailed Reviewers

--- CLUSTER 3: 2 customers (0.2%) ---

SATISFACTION METRICS:

Average Rating: 4.68 ± 0.45

High Ratings (4-5): 100.0%

Low Ratings (1-2): 0.0%

ENGAGEMENT METRICS:

Average Helpfulness: 0.0

Average Review Length: 19734 characters

Average Word Count: 3600 words

TOP PRODUCTS:

1. Amazon Fire TV: 2 reviews (100.0%)

SEGMENT TYPE: BRAND ADVOCATES

MARKETING FOCUS: Leverage for testimonials, referrals, premium products

ENGAGEMENT: Low Engagement, Detailed Reviewers

CLUSTER 4 STRATEGY (25 customers):

RETENTION PROGRAM:

1.Proactive customer service outreach

2.Offer discounts

3.Collect detailed feedback

Focus Product: Amazon Tap - Alexa-Enabled Portable Bluetooth Speaker

Campaign Priority: Medium

CLUSTER 5 STRATEGY (156 customers):

GROWTH PROGRAM:

1.Send satisfaction surveys

2.Offer product recommendations

3.Implement loyalty points system

4.Cross-sell complementary products

Focus Product: Amazon Tap - Alexa-Enabled Portable Bluetooth Speaker

Campaign Priority: High

CLUSTER 6 STRATEGY (9 customers):

RETENTION PROGRAM:

1.Proactive customer service outreach

2.Offer discounts

3.Collect detailed feedback

Focus Product: Moshi Anti-Glare No Bubble Screen Protector for the Fire Phone

Campaign Priority: Medium

CLUSTER 7 STRATEGY (50 customers):

ADVOCACY PROGRAM:

- 1.Recruit for customer testimonials
- 2.Implement referral rewards
- 3.Offer early access to new products
- 4.Create VIP customer tier

Focus Product: Amazon Tap - Alexa-Enabled Portable Bluetooth Speaker

Campaign Priority: High

CLUSTER 8 STRATEGY (1 customers):

RETENTION PROGRAM:

- 1.Proactive customer service outreach
- 2.Offer discounts
- 3.Collect detailed feedback

Focus Product: Amazon 5W USB Official OEM Charger and Power Adapter for Fire Tablets and Kindle eReaders

Campaign Priority: Medium

FUTURE SCOPE :

- 1.Integrate sentiment analysis via NLP
- 2.Real time clustering pipeline for new reviews.
- 3.Personalize offers based on cluster behaviour.

LIMITATIONS :

1. No textual sentiment analysis.
2. Missing values led to data loss.
3. Limited to Amazon device-related products only.

CONCLUSION :

The project successfully demonstrated the unsupervised machine learning methodology in deriving actionable insights from not so perfect customer feedback. Focused on Amazon product review, we transformed raw textual and numerical data into meaningful segments through feature engineering and clustering for market analysis.

The primary objective was to correlate some patterns in customer behaviour which cannot just be available through review counts or average ratings. Using Kmeans we were able to identify nine distinct customer segments with unique review patterns, engagement levels and sentiment orientations.

The evaluation scores demonstrated that the clustering structure was statistically valid and meaningful. Additionally visualization tools confirmed the clusters were well separated and behaviorally diverse.

From a marketing standpoint as well this segmentation provides valuable guidance.

This project bridges the gap between data science and decision making enabling smarter, data-driven marketing strategies grounded in customer behaviour rather than assumptions.

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